

CSCI 7350 Data mining

Homework 3

Wasim Raja Mondal

1. Report the baseline system's average correlation on the test data.

We have used MovieLens 100K dataset.

We find that the baseline recommendation system based on user-based collaborative filtering with Pearson similarity achieved an average Pearson correlation of 0.4318 on the test set.

The evaluation was performed on 200 randomly selected users with random seed 42, and all 200 users produced valid correlation values.

2. For recommendationA.py, explain the modification (e.g., why top-K neighbors, why a minimum shared movie threshold).

In recommendationA.py, the baseline user-based collaborative filtering method was improved by introducing several modifications to enhance prediction reliability and reduce noise.

First, instead of using all users with positive similarity, only the top-K most similar neighbors ($K = 20$) were considered, which helps focus on the most relevant users and reduces the impact of weak or noisy similarities.

Second, a minimum shared movie threshold ($\text{min_overlap} = 5$) was applied when computing Pearson similarity, ensuring that similarity is calculated only when users have sufficient common ratings, thereby improving its statistical reliability.

Third, a mean-centered prediction approach was used, where each user's average rating is taken into account, allowing the model to correct for individual rating biases and produce more balanced predictions. Additionally, only positively correlated users were included in the prediction process to avoid incorporating conflicting preferences.

Together, these modifications aim to improve the stability and accuracy of the recommendation system compared to the baseline approach.

3. For recommendationB.py, explain the item-based approach and why comparing movies across users might improve stability.

In recommendationB.py, an item-based collaborative filtering approach was implemented, where similarities are computed between movies rather than between users.

For a given target movie, the system identifies other movies that are similar based on how users have rated them, and then predicts the rating by using the target user's own ratings on these similar movies. Specifically, adjusted cosine similarity was used, where ratings are mean-centered by user averages to account for user-specific biases. Only the top-K most similar items ($K = 20$) with a minimum number of shared users ($\text{min_overlap} = 5$) were considered for prediction, ensuring that similarity is both strong and statistically reliable. This item-based approach improves stability because movie similarities tend to be more consistent across users compared to user-user similarities, which can be sparse and noisy due to varying user behavior.

As a result, item-based methods often produce more robust and scalable recommendations, especially in datasets with many users but relatively stable item relationships. In this project, the item-based system achieved a higher average Pearson correlation of 0.5314, outperforming both the baseline and the modified user-based approach, demonstrating its effectiveness and stability.

4. Report the average correlation values for recommendationA.py and recommendationB.py, comparing them with the baseline.

The performance of the three recommendation systems was evaluated using the average Pearson correlation between predicted and actual ratings on 200 test users. The baseline user-based collaborative filtering system achieved an average correlation of 0.4318. The modified user-based system implemented in recommendationA.py achieved an average correlation of 0.3722, which is lower than the baseline, indicating that the applied constraints (top-K neighbors and minimum overlap) may have reduced useful information in this case.

In contrast, the item-based collaborative filtering system implemented in recommendationB.py achieved the highest performance, with an average correlation of 0.5314, significantly outperforming both the baseline and recommendationA. This result demonstrates that item-based approaches provide more stable and accurate recommendations, likely due to more consistent similarity relationships between movies compared to users.

5. Describe any other improvements made to the programs.

In addition to Pearson correlation, ranking-based evaluation metrics were also computed for recommendationA.py to further assess recommendation quality. Relevant movies were defined as those with true ratings greater than or equal to 4, and the evaluation was performed using a cutoff value of $K = 10$. Based on this definition, the system achieved a **Recall@10 of 0.6293** and a **MAP@10 of 0.7251**. These results indicate that the modified user-based collaborative filtering approach is effective at retrieving relevant movies within the top-10 recommendations and ranking them in higher positions. While the correlation metric evaluates prediction accuracy, these ranking metrics provide additional insight into the practical usefulness of the recommendation system in identifying and prioritizing relevant items.